

10 Microtubules as a Quantum Hopfield Network

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Summary. Penrose and Hameroff’s orchestrated objective (Orch OR) theory [14] suggests the existence of quantum computation in microtubule protein assemblies inside living cells. **To investigate this suggestion numerically, we model this system as a quantum Hopfield network (QHN) with qubits representing the tubulins and interacting coulombically with each other, at finite temperatures.** The effects of energy losses, dissipation, and other environmental factors are not considered, or are considered to be screened. Simulations are carried out on this computational model, and we look for stable states (local minima) of the network. We find that quantum information processing in microtubules is feasible, though at temperatures (5.8 K) lower than physiological temperatures.

10.1 Introduction

There are two major architectures for computation: the centralized algorithmic, and the decentralized adaptive. Most everyday machine computing uses the former model, which is enormously successful in a number of applications. Everyday biological computing uses the latter. There are also a number of tasks that are less suited to the algorithmic approach: for example, pattern recognition, or the traveling salesman problem [12]. It can also happen, for a number of reasons, that an algorithm is very difficult to develop for a particular problem.

In fact, quantum computing in general may fall into this category [4, 5]. Despite now many years of intensive study by a large number of well-funded and very bright people, the number of truly quantum algorithms known is very small. We do not even know what problems it would be possible to solve were we to build a large-scale quantum computer, much less how to write algorithms to solve them. The full power of quantum computing may never be realized unless we discover a means of answering both kinds of questions.

In previous work, we have suggested the use of quantum neural computers to address at least the latter need [4, 1, 5]. An adaptive or neural computer in a sense constructs its own algorithm for the problem given from the training set supplied. (In addition, neural computers may be the answer to the extremely vexing problems [7] associated with error correction in quantum computers.)

Our own brains are clearly of the decentralized, adaptive type. Could they also be quantum mechanical, not just in their underlying chemical processes, but also in their processing of information? Hameroff and Penrose [14] have suggested that the process of human consciousness cannot be simulated classically. They have proposed an orchestrated objective reduction (Orch OR) model in which quantum computation takes place in the microtubule protein assemblies present in the neurons of the brain. On the face of it this seems unlikely: decoherence at physiological temperatures, one would guess, would be rapid. But it may be true that biology, as Hagan et al. suggest, “can take advantage of quantum modes in clever ways” [10] by building mechanisms to maintain coherence, even in the “warm, wet, and noisy” environment of the living brain.

In this chapter we take a step towards the exploration of this question, by simulating a quantum neural network with the structure of a microtubule. We use an extremely simplified model [18] of the tubulin dimers: each is represented simply as a qubit, a single quantum two-state system, the quantum analog of a classical bit. A qubit, unlike a bit, can exist in a superposition of the two states. We assume that environmental factors are negligible: for example, that the thermal fluctuations of the bath are screened by a Debye layer of counterions, as suggested in Hagan [10]. We do include Coulombic interactions between qubits, and thermal effects for the qubits themselves. This is a kind of quantum Hopfield network [4, 11]: Each of the individual processing elements or neurons (here, the qubits) interacts with each of the others (a fully interconnected network); the elements (qubits) are initialized in some state, and allowed to evolve towards a local minimum. These networks store information and are also called “associative memory” or “content addressable memory” models. The final stable state is a pattern that is recalled by the network. We will look for evidence of locally stable states that represent quantum information processing.

10.2 Microtubulin Model

Real biological neurons have a highly ordered internal network known as the cytoskeleton, a protein polymer that is responsible for cell movement and organization of the organelles within the cell. The cytoskeleton consists of microfilaments, microtubules, actin filaments, and intermediate filaments. Of these components, microtubules display the greatest structural versatility [17] by determining the cell shape and providing a set of tracks for the movement of cell organelles. Microtubules are conveyor belts inside the cells. With the help of special attachment proteins (MAPs), they move vesicles, granules, organelles like mitochondria and chromosomes. They form the skeleton (cytoskeleton) that gives shape to the cell and help in the transportation of materials inside the cell. The microtubules are linked to other microtubules through microtubule-associated proteins (MAPs). Microtubules

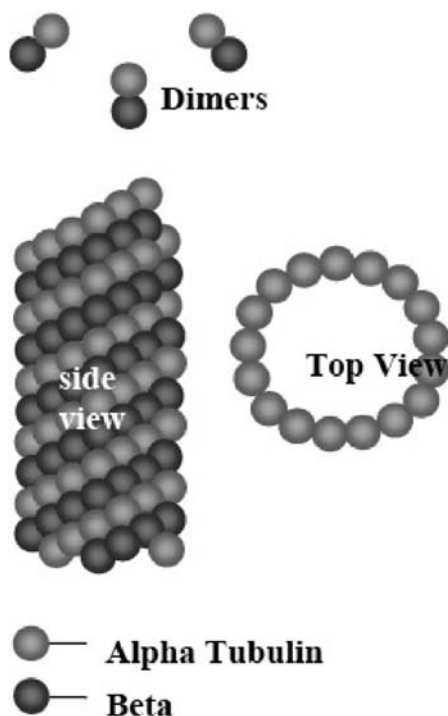


Fig. 10.1. Structure of a microtubule and the arrangement of its tubulins

are long, hollow cylinders that are 25 nm in diameter and their length may extend to macroscopic dimensions. They are composed of many smaller double molecules known as tubulin dimers. Fig. 10.1 shows the structure of a microtubule and the arrangement of its tubulins. The microtubules usually have 13 tubulins in circumference though there can be 12–16 tubulins depending on the environment, mainly ion concentrations. The tubulin dimers are arranged in a skewed hexagonal lattice. Each tubulin dimer is made up of two similar tubulin monomers called alpha tubulin and beta tubulin having slightly different properties. The tubulin dimers are 8 nm long and the monomers are 4 nm long with a molecular weight of 55 kDa. The dimers bind calcium ions and have a net electronegative charge due to the presence of acidic amino acids. This electronegative charge is predominant towards the alpha monomer that creates an oriented dipole in the tubulin dimer. The monomers bind together to form the microtubules.

Because each tubulin acts as an oriented dipole, the microtubule as a whole is an assembly of oriented dipoles. This is a kind of two-level system (TLS). The tubulins interact coulombically with their neighbors, and such a system exhibits some frustration in the ground state as it cannot satisfy all the dipole couplings with its neighbors. Couplings are created

due to the charge transfer within the dimers and their associated conformational changes. This coupling leads to the presence of piezoelectric properties that are very important in microtubule signaling, communication and assembly/disassembly behavior. Some experiments and simulations have shown that the microtubules can perform classical information processing.

Some of the functions of microtubules like assembly/disassembly, signaling and information processing could be related to the ground-state properties of the lattice of the dipoles [17]. Three kinds of classical arrangements of the dipoles are possible: (1) random arrangement where there is no order in the dipole arrangements; (2) ferroelectric arrangement, which is a long-range parallel-aligned order; and (3) intermediate or weak ferroelectric arrangement, where a short-range order is present. These three arrangements may exist under different conditions of temperature, electric field and microtubule length. A microtubule can switch among these arrangements. For assembly/disassembly and signaling processes, microtubules use a long-range ferroelectric arrangement, and a short-range weak ferroelectric arrangement for information processing. Are these arrangements influenced or perhaps produced by quantum effects?

10.3 Hopfield Model

The Hopfield neural network is a model of an associative content-addressable memory; it is capable of storing information as well as carrying out certain computational tasks such as error correction. In Hopfield neural networks the processing units, or neurons, are fully interconnected, i.e. each unit is connected to every other unit. Hopfield [11] demonstrated that an associative memory can be implemented with his network and hard optimization tasks such as the traveling salesman problem can be solved. A Hopfield network is initialized to a beginning state. Then each neuron's output is updated according to its weighted inputs. The updating of the neurons continues until the network converges to a stable state, or fixed point.

A *binary* Hopfield network has neurons that can be in one of two states: $S_i = +1$ (on) and $S_i = -1$ (off), where i is the number of the neuron. These are exactly analogous to bits 0 and 1. Each neuron is synaptically connected to other neurons with multiplicative weights that are real numbers. Each neuron is randomly selected and sums up its input and decides whether to turn itself on or off. Let W_{ij} be the synaptic strength from neuron ' j ' to neuron ' i ' and let I_i be the external input to unit ' i '. Each neuron takes a weighted sum of its inputs according to the following equation:

$$S_i = \text{sgn} \left(\sum_{j \neq i=1}^n W_{ij} S_j + I_i \right). \quad (10.1)$$

After repeated updating, the network may achieve a stable point, meaning, in the state space, all neurons remain in their current state after examining the inputs. The output is a stable vector-of-states that constitutes the recall of a stored pattern in the memory.

In the *continuous* Hopfield net, output values can be in the interval $(-1, +1)$. The governing equation of the i th neuron is

$$\begin{aligned}\frac{du_i}{dt} &= \sum_j W_{ij} S_j - \frac{u_i}{R_i} + I_i, \\ S_j &= \tanh(\gamma u_i) = f(u_j),\end{aligned}\tag{10.2}$$

where R_i is the decay rate ($R_i > 0$), S_j is the output of unit j after the activation function f is applied, and γ is the gain of the activation function. The Hopfield network considered in this chapter is in the high-gain limit where γ and R_i are large, and the $\tanh$ approximates the sgn function of the discrete network.

Each state in the high-gain Hopfield neural network has a Lyapunov function V , given by [13]

$$\begin{aligned}V &= L(\{S_i\}) - L(\{S_i^*\}), \quad \text{where} \\ L &= -\frac{1}{2} \sum_i \sum_j W_{ij} S_j S_i - \sum_i S_j I_j.\end{aligned}\tag{10.3}$$

Here, L is a function of the states of all the neurons of the system $\{S_i\}$, and $\{S_i^*\}$ is the state of the system for which L is a minimum. V is a Lyapunov function for the dynamics of (10.2) since it is positive near the local stable state $\{S_i^*\}$, and it can be shown [11] that $dV/dt < 0$ except at the fixed points. Thus the Hopfield network is globally asymptotically stable and the stable states are the minima of the L function of the system. The processing of the Hopfield neural network, from an initial to a final steady state, is a process by which the L function of the system becomes smaller and eventually achieves a local minimum at the stable state. The weights, the external inputs, and the initial state determine which stable state, or pattern, is recalled by the network.

10.4 Quantum Model

The quantum system can be put into the Hopfield form to obtain the quantum Hopfield net (QHN) as follows. Consider an array of N qubits. We write the Hamiltonian, or energy function, for each qubit j as

$$H_j = K\sigma_{xj} + A\sigma_{zj},\tag{10.4}$$

where $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ and $\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ are the Pauli matrices, and the full Hamiltonian $H = \sum H_j$. The first term represents the flipping of the qubit from one state to the other, with amplitude K . This term allows a quantum system to switch from one state to another without having the energy to climb the associated energy barrier and is therefore called “tunneling”. The effect of σ_x can be understood by looking at its operation on a state $\begin{pmatrix} \delta \\ \gamma \end{pmatrix}$. This state has probability amplitude δ of being found in the +1 state, $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$, and γ of being found in the -1 state, $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$; σ_x operating on this state gives us $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \delta \\ \gamma \end{pmatrix} = \begin{pmatrix} \gamma \\ \delta \end{pmatrix}$, in which the amplitudes are flipped. In particular, the +1 state is flipped to the -1 state, and vice versa. The second term in (10.4) represents the energy difference $2A$ between the two states. This difference can be the result of external fields or interaction with other qubits. The Pauli matrix σ_z measures the state of the system; for example, $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = (-1) \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, telling us that the system is in the -1 state. The difference between the operation of σ_z on the +1 state and on the -1 state is $(+1 - (-1)) = 2$, so $A\sigma_z$ gives us an energy difference of $2A$ between the two states.

For the whole system we need to include all the qubits and their interactions. For a system at finite temperature we write the partition function $Q = \text{tr} e^{-\beta H}$, where tr is the trace of the matrix and β the inverse temperature in units of Boltzmann’s constant. The partition function contains information about all possible states of the quantum system, and how they depend functionally on each variable and each parameter. This will lead us to our definition of the quantum Hopfield net. Following [3] and [1], we write the trace in the basis set of state variables $\{S_j = \pm 1\}$, and use the Feynman path integral [8] formulation:

$$Q = \int \prod_{j=1}^N D[\sigma_{zj}(t)] \exp \left\{ - \int_0^\beta dt \sum_{j=1}^N K \sigma_{xj}(t) + \int_0^\beta dt \int_0^\beta dt' \sum_{j \neq j'=1}^N \sigma_{zj}(t) \alpha_{jj'}(t, t') \sigma_{zj'}(t') \right\}, \quad (10.5)$$

The path-integral formulation is one way of writing the trace: as a summation over all possible paths $\sigma_z(t)$ for each of the j qubits. Here, α is the Coulombic interaction between all the pairs of qubits of the array. The first term in (10.5) is the tunneling term that represents the probability amplitude that the qubit can make transition from one state to another. The second

term represents the coulombic interaction between the qubits. (10.5) can be thought of as propagation in imaginary time, because the Boltzmann factor $\exp\{-\beta H\}$ is the same as the expression for time evolution $\exp\{-i t H/\hbar\}$ for $t = -i\beta\hbar$, where $\hbar$ is Planck's constant divided by 2π .

To make progress in evaluating the path integral, we discretize the qubits of the QHN. This transforms the path integral over the continuous states of the N qubits, into a set of sums over states of the N qubits at n discrete times. Thus we can, for finite n , compute the state of the net by summing over all possible states of each of the discretized points of each of the qubits. If we let $n \rightarrow \infty$, the discretization is exact. The approximation of finite n will be good as long as $\beta H/n$ is small. Because the partition function is a trace, the state of each qubit at "time" $t = 0$ is constrained to be the same as its state at "time" $t = -i\beta H$. Thus we can picture each qubit as a loop, having n discretization points, propagating in imaginary time from 0 to β . A discretization point at any instant is the instantaneous state of the qubit at that corresponding value of inverse temperature. Figure 10.2 shows a qubit with 6 discretization points. While the loop is here drawn as a circle, it should not be imagined as being constrained so: all possible paths from the starting point to the ending point should be imagined as being included in the sum. The lines between discretization points represent the tunneling term in the Hamiltonian. (The analogous picture for a continuum position variable x is perhaps easier to understand; see, e.g., [6].) Qualitatively, we can think of the size of K , the tunneling amplitude, as indicating the floppiness of the bonds between adjacent discretization points: the larger it is, the less constrained is one point by its immediate neighbors, and thus, the more likely is the instantaneous state of the system at an intermediate imaginary time to flip from one state to the next.

Physically, the quantum-mechanical nature of the network increases as the discretization number n increases. In the picture, we can think of the amount of phase space (different states) the system is free to explore, as increasing when we increase the number of discretization points. That is, the loop can become floppier and floppier, for a given value of K , if the number of intermediate intervals is increased. On the other hand, if n is reduced, the loop will have less and less freedom to explore different states, and if we set $n = 1$ the loop must shrink to a single point, with no freedom to flip its state at all. This is the classical situation: no quantum-mechanical tunneling.

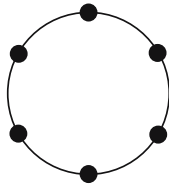


Fig. 10.2. A qubit with 6 discretization points

We can also think of this same situation in terms of the inverse temperature. The inverse temperature β is the total “length” of the path the qubit has to travel, from its initial point out into the world and then back. At low temperature, which is the quantum limit, β is large, which means the system can explore a large amount of the phase space (and we need to set n large in order to have a good approximation); at high temperature, $\beta \rightarrow 0$ and the loop necessarily shrinks to a point (the classical limit.)

Rewriting the path integral in terms of discretized sums, (10.5) becomes:

$$Q = \lim_{n \rightarrow \infty} \sum_{S_j = \pm 1} \exp \left\{ \zeta \sum_{j=1}^N \sum_{t=0}^{n-1} S_j(t) S_j(t+1) + \sum_{j \neq j'-1}^N \sum_{t=0}^{n-1} S_j(t) \alpha_{jj'}(t, t') S_{j'}(t') \right\}, \quad (10.6)$$

Here, ζ is the tunneling term that is given by the expression $\zeta = -\frac{1}{2} \ln[\tanh(\beta K/n)]$. Again, we use the variable t to label the inverse temperature ranging from 0 to β , but it is now a discrete variable, ranging from 0 to $n-1$. The functions $\{S_j\}$ refer to the instantaneous values assumed by each of the qubits j ; i.e. the eigenvalues of the operators $\sigma_z(t)$ for each of the j qubits, frequently called “spins” in the literature. Thus, $\{S_j(t)\}$ is a microstate of the system; and the summation over all the microstates $\sum_{S_j = \pm 1}$ gives the partition function. The contribution of microstates to the partition function is large as their associated energy is low. For an infinite discretization value ($n \rightarrow \infty$), the discretization representation is exact, as indicated by the equality sign in (10.6) and as noted above, but since we do not have an infinite (or quantum) computer on which to do the simulation, we will make do with finite n . As long as $\beta H/n$ is much less than one, the discretization error should be small. Only infinite n and $n = 1$ have recoverable physical meaning, as corresponding to the quantum and classical cases, respectively; however, as we increase n we can investigate the role of increasing quantum-mechanical character of the network.

10.5 Quantum Hopfield Network

A quantum Hopfield network (QHN) has an enormously expanded range of states as the qubits can exist in any superposition of the two states. This is not the same as a continuum classical net, since the state of the system is not between the two states, but both states at the same time: the system carries information on as many channels as there are states contributing to the superposition. Under some circumstances that information is accessible, as in the operation of some quantum algorithms [16, 9].

Recall that the contribution of microstates to the partition function is large if their associated energy is low. In (10.6), the states that are more probable could be obtained by minimizing the argument of the exponential. We rearrange that argument to correspond with the L function for a Hopfield network. That is, we define the L function for the QHN as the integral of the Hamiltonian over the imaginary time loop, divided by β , i. e. the quantity in curly brackets in (10.6):

$$L_{\text{QHN}} = -\frac{1}{\beta} \left[\zeta \sum_{j=1}^N \sum_{t=0}^{n-1} S_j(t) S_j(t+1) + \sum_{j \neq j'=1}^N \sum_{t=0}^{n-1} S_j(t) \alpha_{jj'}(t, t') S_{j'}(t') \right] \quad (10.7)$$

This is the QHN. Classically, the high-gain Hopfield net replaces a continuum of possible values of each neuron with a finite number; in the QHN it becomes again a continuum, as described below. Information in the Hopfield net is stored as stable states. Classically the discrete net converges to the corners of an N -dimensional box [11], where the neuron outputs necessarily equal plus or minus 1. In a network of N classical neurons there are 2^N possible states (corners of the box.) Once we discretize the quantum Hopfield net to discretization number n , there are $2Nn$ possible states: that is, each corner of the N -dimensional classical Hopfield box has $2n$ states. The greater the discretization n , the greater the number of possible states. When the temperature is lower we need a larger discretization number; thus, the lower the temperature the greater the storage capacity of the QHN. For infinite n we have the fully quantum-mechanical system, and have a fully faithful representation of the quantum dispersion – all possible superposition states.

To see how our quantum loop picture works with the Hopfield formulation, consider now the discretized path integral picture for more than one qubit, a QHN at finite n . Figure 10.3 shows a picture of the net for two qubits ($N = 2$) and two discretization points ($n = 2$). Each loop represents a qubit or two-level system (TLS), propagating in imaginary time (inverse temperature), from $t = 0$ to $t = \beta$. Each dot on a loop represents the instantaneous state of the qubit at a particular value of t , and these dots interact with their nearest neighbors along the loop (the first term in (10.6) or (10.7)). This is the tunneling term. The arrows show the direction of integration, from $t = 0$ to $t = \beta$. Each dot also interacts with other dots on other loops according to the second term in (10.6). In the simplest possible model, which we consider here, the two qubits interact only at equal imaginary times t , and the Lyapunov function from (10.7) becomes:

$$L_{\text{QHN}} = -\frac{1}{\beta} \left[\zeta \left(\sum_{i=1}^n S_i S_{i+1} + \sum_{i=1}^n S'_i S'_{i+1} \right) + \alpha \left(\sum S_i S'_i \right) \right], \quad (10.8)$$

where the (primed, unprimed) S functions refer to the two interacting loops.

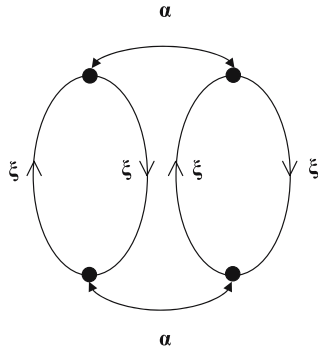


Fig. 10.3. Representation of two interacting qubits, with discretization $n = 2$. Each loop represents a qubit, propagating in imaginary time (inverse temperature) from 0 to β . The interactions between discretization points on the same loop have a strength of ξ ; the (bidirectional) interactions between points on different loops, of α

10.6 QHN as an Information Propagator for a Microtubules Architecture

The microtubule structure is cylindrical in shape consisting of a number of tubulins arranged in a hexagonal lattice structure, as shown in Fig. 10.1. The tubulin dimers are oriented as they have a net electronegative charge towards the alpha-monomer. There are 13 tubulins along the circumference of the microtubule. Each tubulin has six neighbors and interacts with its six neighbors. There can be propagation of information from one qubit to another in the network due to the (coulombic) interactions between each pair of qubits. The output is calculated as the lowest energy state of the qubit. Figure 10.4 shows the dipole–dipole interactions of a single dot with its neighbors, where a '+' sign represents a parallel arrangement and a '-' sign represents an antiparallel arrangement of the dipole moments.

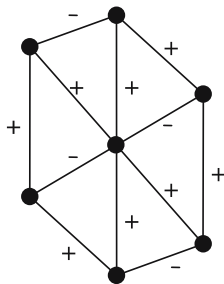


Fig. 10.4. Dipole–dipole interactions showing parallel or antiparallel arrangements of dipole moments

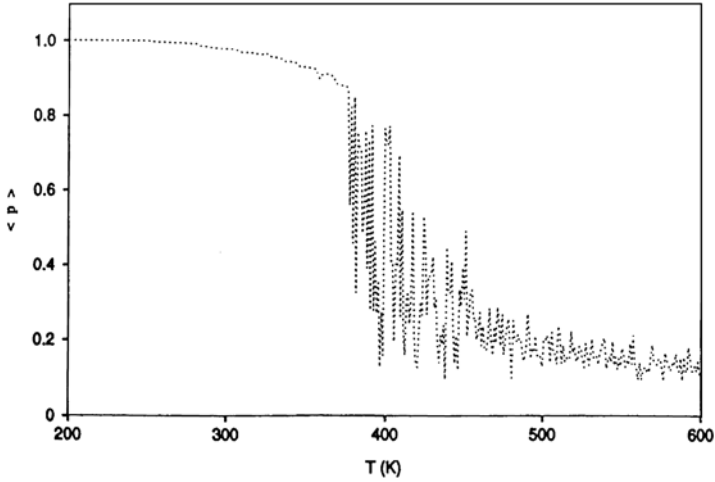


Fig. 10.5. Mean polarization per layer, as a function of temperature, for a 13×26 lattice, from [18], as calculated by classical Monte Carlo

We now apply our QHN methodology to the array of microtubules. The values of intra- and intermolecular distances, length of microtubule (based on the number of rows), tunneling amplitude, temperature, external field, leakage term, gain parameter, etc., are set, using experimental values where possible. The QHN is started out at some initial state after assigning values to all the parameters in the algorithm, by assigning an initial dipole value (orientation of the polarization) to each qubit. The output of each neuron is updated according to its weighted inputs. The updating procedure of the network minimizes the Hopfield energy or Lyapunov function and continues until the network converges to a stable state achieving a local minimum. This stable state is a pattern that is recalled by the network. This final state is the output of the network.

To the test the validity of our method we first compare it to some published classical calculations on this model. Tuszynski, et al. [18], have performed some Monte Carlo computations to observe the size dependence of the phase behavior in microtubules. Figs. 10.5 and 10.7 are reproduced from that chapter. Figs. 10.5 and 10.6 show the mean polarization per site for a triangular lattice whose size is 13×26 (26 layers, 13 qubits in each row, 338 qubits), in the classical limit ($n = 1$). The initial values of spins Fig. 10.6 were taken from [18]. It can be seen that the QHN results are comparable to the Monte Carlo calculations. For the large-lattice limit, Tuszynski has a calculation of 5000 layers; due to computational capacity problems we were only able to do 3000 layers; nonetheless our results are comparable to theirs, as seen in Figs. 10.7 and 10.8. Again, Figs. 10.7 and 10.8 are in the classical limit.

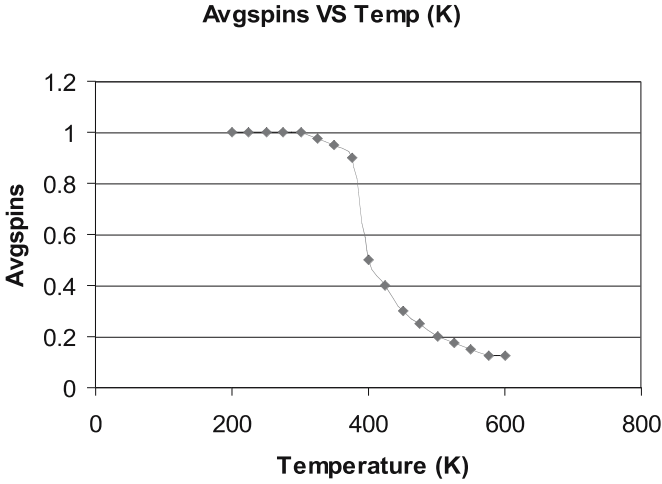


Fig. 10.6. Average polarization as a function of temperature, for a 13×26 lattice, using values for the parameters of [18], calculated by QHN, in the classical limit ($n = 1$)

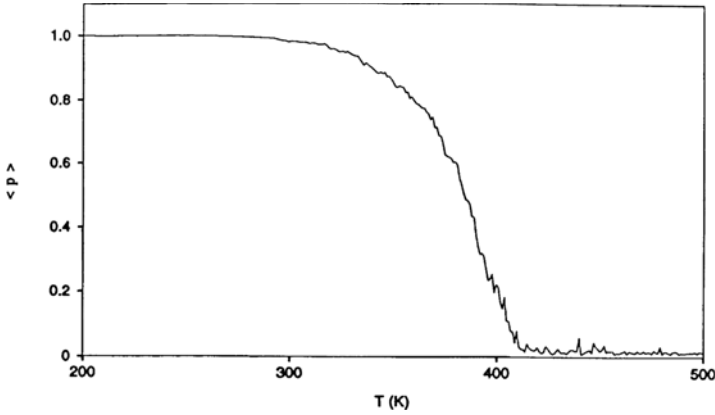


Fig. 10.7. The mean polarization for a 13×5000 lattice, as calculated by classical Monte Carlo [18]

With some confidence in our method we now move to nonzero quantum-mechanical effects, setting $n > 1$. In order that our discretization be valid we require $\beta E/n < 1$, where E is the energy of the system. Because of computational limitations we used only a 26-layer system, or 338 qubits. All other experimental parameters were kept as before, and we look for local minima for

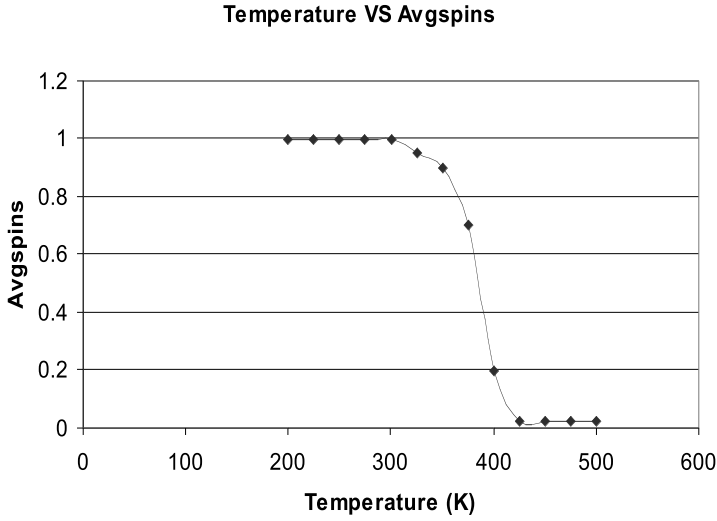


Fig. 10.8. Average polarization as a function of temperature for a 13×3000 lattice, calculated by QHN, in the classical limit ($n = 1$), using the parameters of [18]

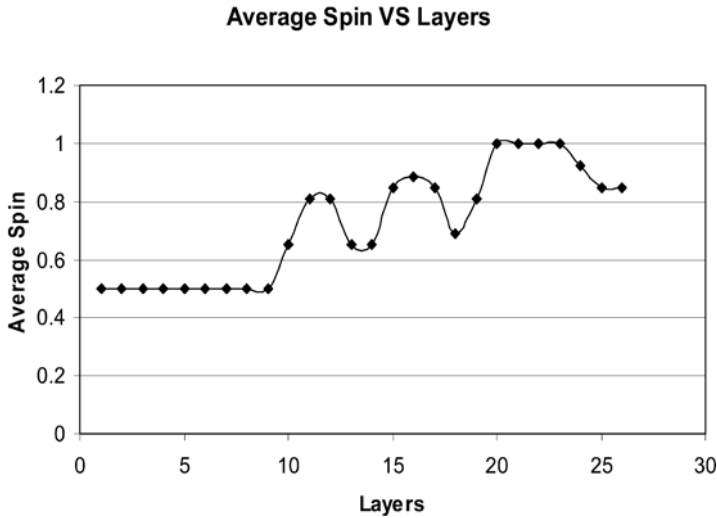


Fig. 10.9. The variation of average spins along the layers of the 13×26 microtubule for $n = 4$ with an initial configuration of 1 down, 3 up spins for each qubit, as calculated by QHN, at $T = 5.8$ K. This final configuration was stable at 100 000 passes

the QHN. Some interesting phenomena were observed. An example is shown in Fig. 10.9, where we see the variation of average spin along the layers of the microtubule, at a temperature of 5.8 K ($\beta = 2$ MeV), starting from an

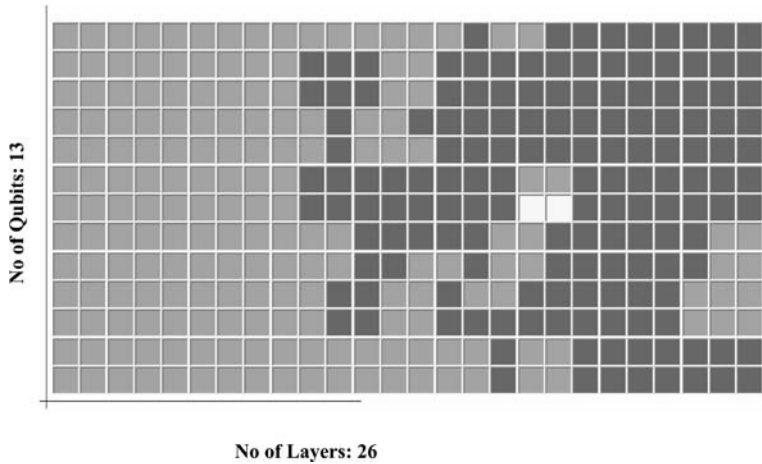


Fig. 10.10. Spin configuration, by qubit, of the system in Fig. 10.9

initialization of the qubits in a superposition state of 1 down and 3 up spins. The QHN was stable at 100 000 epochs.

It can be observed from the above graph that there is a locally stable configuration in which a nearly constant polarization at one end is transformed to a different one at the other, a kind of information processing effect. That is, we have found a stable configuration for which the “input” at one end of the microtubule is transformed to an “output” at the other, and both input and output are superposition states (i.e. the average spin (polarization) is between minus one and one). Since superposition states cannot be realized classically, the microtubule is acting as a quantum computer, with fully quantum-mechanical inputs and outputs. This is of course lower than physiological temperatures, and indeed no similar “processing” effect was observed at temperatures that high: the higher the temperature, the smaller the quantum loop, which represents the quantum dispersion, shrinks. At physiological temperatures the loop is essentially a single point, and no quantum effects (which are due to the spreading out of the loop) can be observed.

Figure 10.9 can be better understood by looking at each qubit, rather than only layer averages. In Fig. 10.10, each cell represents a qubit and its color represents its average spin value. The spin values vary from -1 to $+1$. The color for $+1$ is red, the color for 0.5 is orange, and the color for 1 is light yellow. There is a variation of color from red to light yellow as the spin value changes from $+1$ to -1 . So in fact though the average spin along a layer varies continuously from $+1$ to -1 , no single spin has any value other than $+1$, 0.5 , or -1 ; these seem to be the only stable values for the superposition states for these conditions. This may be an artifact of the discretization level: with only four discretization points per qubit there are only five possible distinguishable superposition states (1 , 0.5 , 0 , -0.5 , and 1 , which can be produced in 1 , 4 ,

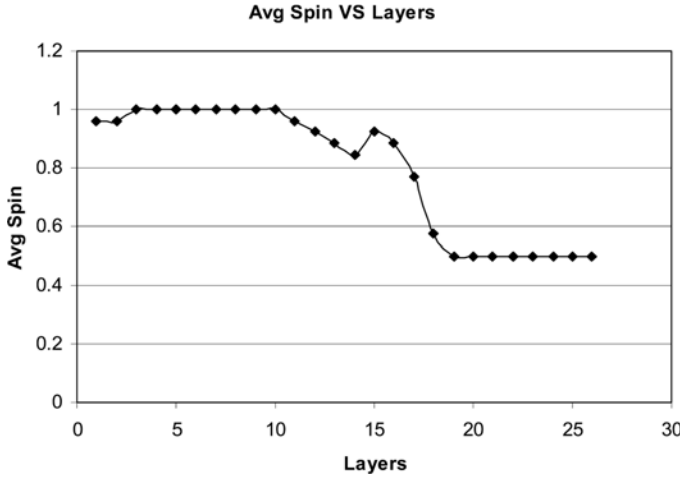


Fig. 10.11. The variation of spins along the layers of the microtubule for $n = 4$ where the initial spin configuration is the reversed final spins of the system whose initial configuration is 1 down, 3 up spins, as calculated by QHN, at $T = 5.8$ K. This final configuration was stable at 100,000 passes

6, 4, and 1 ways, respectively.) Different individual runs do produce different final averages for individual qubits; however, the overall pattern by layers is relatively stable.

Information processing of this type was fairly easy to produce, if the system was initialized in a superposition state of almost any kind. For example, an initial situation reversed from the one that produced Fig. 10.9, gives us a stable configuration like that shown in Fig. 10.11. That is, the initial spins for the qubits were the reversed final spin values obtained from the initial spin configuration of 1 down, 3 up spins for each qubit.

Another stable processing configuration was produced by setting the initial spins for the qubits to the final spin values obtained from the initial spin configuration of 1 down, 3 up spins for each qubit. The stable final configuration is shown in Fig. 10.12. Figure 10.12 is similar to the previous graph (but in the reverse direction) which has reversed final spins as initial spins.

Because our calculations are approximate (finite n) it behooves us to check that the discretization error is negligible. Thus we repeated our calculations for a discretization ten times the size, i.e. $n = 40$. We took the initial spin configuration to be the final spin values of the $n = 4$ simulation (whose initial configuration was 1 spin down, 3 spins up), mapped to $n = 40$ such that the average was the same as for $n = 4$. Figure 10.13 shows the results, which while not identical are very much analogous to those for $n = 4$. We are thus confident that we are observing a real phenomenon, not an artifact of the discretization, at least in the overall information processing effect of the tubulin array.

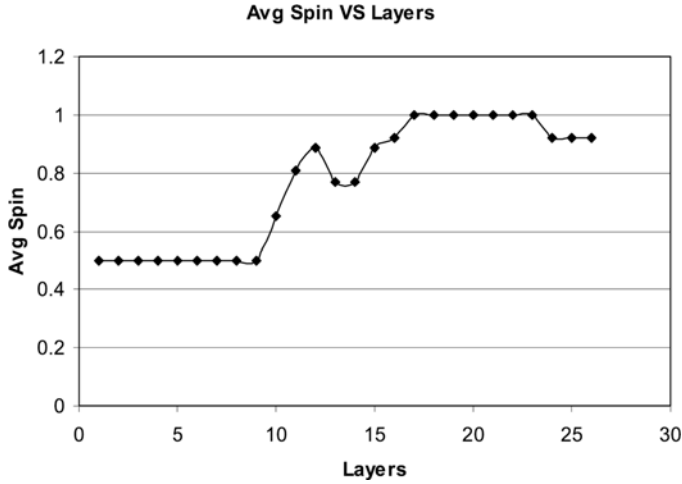


Fig. 10.12. The variation of spins along the layers of the microtubule for $n = 4$ where the initial spin configuration is the final spins of the system whose initial configuration is 1 down, 3 up spins, as calculated by QHN, at $T = 5.8$ K. This final configuration was stable at 100 000 passes

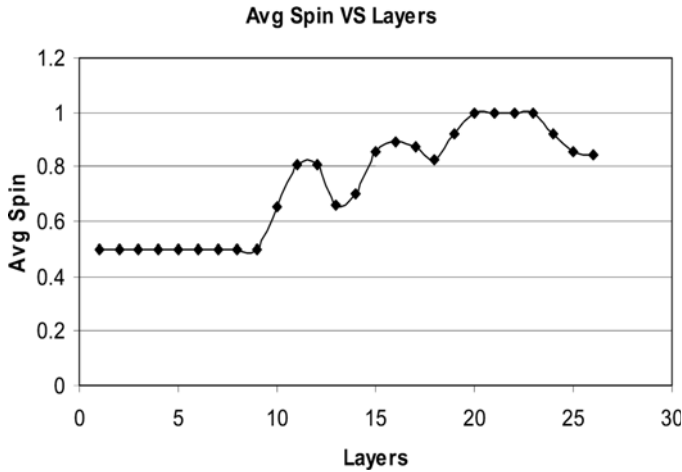


Fig. 10.13. The variation of spins along the layers of the microtubule for $n = 40$

Classical processing of information was even easier to produce. In Figs. 10.14–10.17 we show a number of classical switching configurations, produced with various different initial configurations, as detailed in the figure captions. All but one of the simulations shown stabilized at a classical switch configuration (± 1 at each end). A large number of other configurations were tried. Long-range ferroelectric configurations (no switching) were also easy to stabilize.

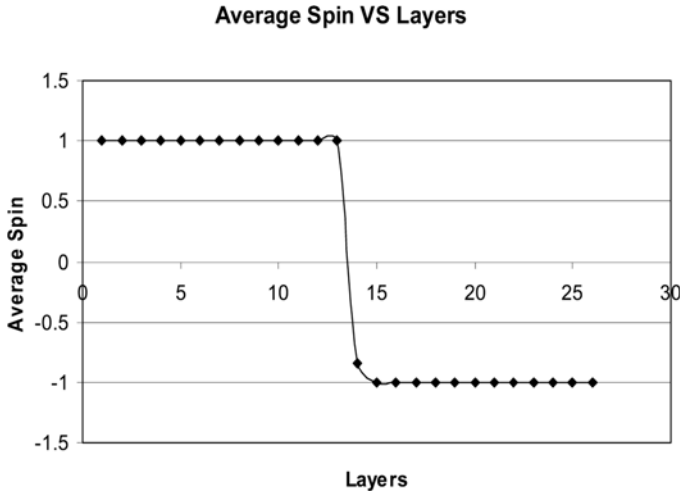


Fig. 10.14. Variation of spins along the layers of a microtubule for $n = 4$, starting from an initial spin configuration where half qubits (169) have all spins up and half qubits (169) have all spins down

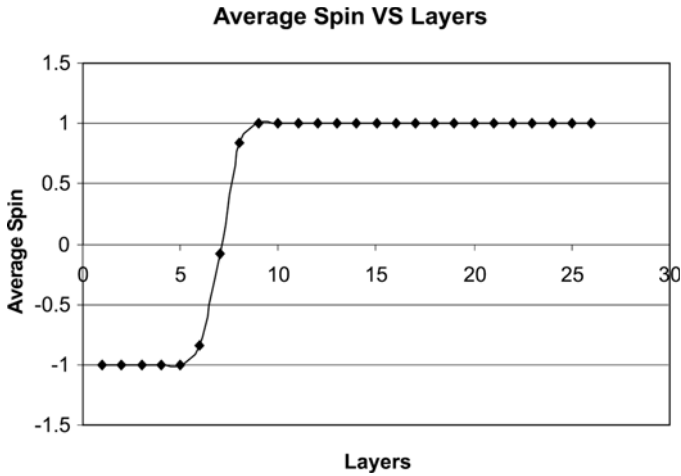


Fig. 10.15. Variation of spins along the layers of a microtubule for $n = 4$ having an initial spin configuration where a quarter of qubits (85) have all spins down and three-quarters of qubits (253) have all spins up

10.7 Conclusions and Future Work

We have shown that a simplified model of a microtubule, in which the tubulin dimers are represented as qubits, interacting with each other coulombically at finite temperature, can in fact exhibit quantum information processing, though at temperatures much lower than physiological temperatures. Still,

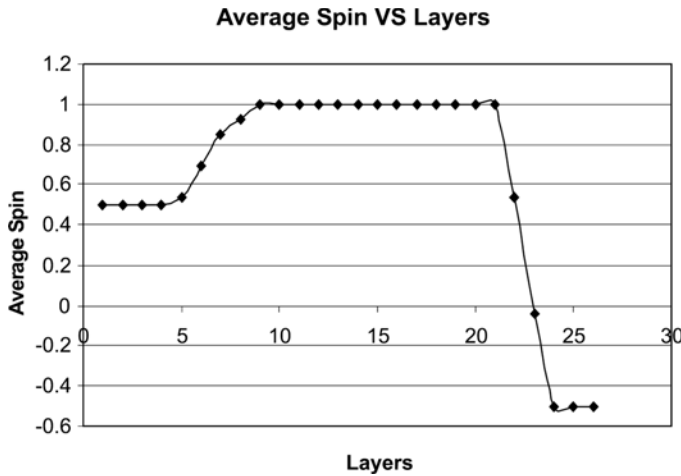


Fig. 10.16. Variation of spins along the layers of a microtubule for $n = 4$ having an initial spin configuration where all qubits in odd layers have 3 spins up and 1 spin down and all qubits in even layers have 3 spins down and 1 spin up

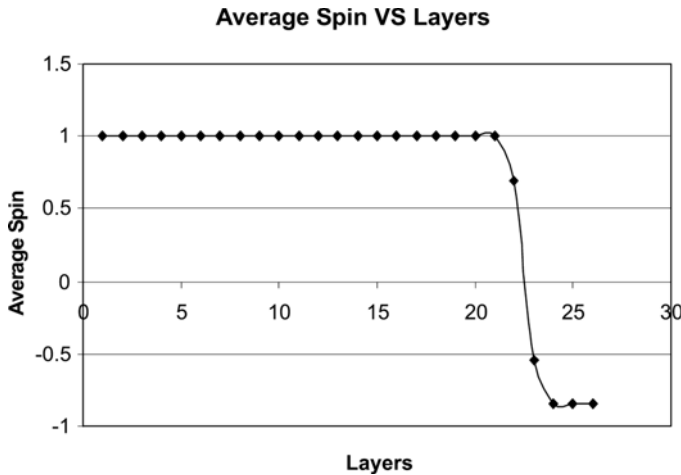


Fig. 10.17. Variation of spins along the layers of a microtubule for $n = 4$ having an initial spin configuration where the first 12 qubits in odd layers have all spins up and the 13th qubit has all spins down. The first 12 qubits in the even layers have all spins down and the 13th qubit has all spins down

all that is necessary for the temperature to be “low” is that it is small compared to the relevant energy scales in the problem; as is well known, quantum information processing has been successfully done at high temperatures [19]. While we have made every effort to ground the parameters for our simulations in experiment, some guesswork was necessary; if we are wrong in places

that could affect the scale at which our observed phenomena occur. We were also able to find stable classical switching configurations, and long-range ferroelectric configurations suitable for signaling or assembly/disassembly. Of course, the fact that quantum conditions can produce observed classical results does not show that those effects have in fact come from the simulated quantum conditions.

We have assumed a rather primitive model, due to computational capacity limitations; in particular, we have assumed most environmental effects to be screened. It would be interesting to do a detailed study to see if this is in fact the case. Another possible objection that could be made is that the simulations found only locally stable configurations: the Hopfield minimization procedure does not mimic the actual dynamics, and the configurations found are not necessarily kinetically accessible.

Another possible approach would be a larger-scale one: to determine what kinds of processing it is that we think a brain is doing, and try to train a quantum network (e. g., this model) to reproduce the training set and to generalize the result. Along those lines, we have developed a method [2] for training a spatial network like this model, based on Werbos's [20] earlier work; we have succeeded in training a small network (six qubits) to solve the controlled-NOT problem. Of much more interest would be the possibility of training the net to perform a truly quantum problem, like the calculation of entanglement; that work is in progress [5].

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